

From a Cubic Hénon Channel to CM Frobenius Data: An Exact Arithmetic Bridge and Its Scope Boundary

Liang Wang^{*1}

¹School of Artificial Intelligence and Automation, Huazhong University of Science and Technology
Wuhan 430074, P.R. China

^{*}Corresponding author

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Abstract

Artificial prime damping makes a three-channel Hénon–Kummer block determinant trace class but adds an external clock. We replace that damping by the minimal geometric carrier of cubic symmetry, the elliptic curve $E : y^2 = x^3 + 1$. Its order-three automorphism gives complex multiplication by $\mathbb{Z}[\zeta_3]$, and its local polynomial is $1 - a_p T + pT^2$. We prove directly that $a_p = 0$ for every good prime $p \equiv 2 \pmod{3}$, because the cube map is bijective in that residue class. An exact certificate counts all good primes below 2,000, verifies every Hasse bound, and independently enumerates $E(\mathbb{F}_{p^2})$ at four primes. This supplies a genuine arithmetic connection and intrinsic square-root normalization. It does not supply a Hénon periodic-orbit determinant: the resulting object is the CM elliptic L -function, with different local and global analytic data from the Riemann ξ -function.

1 Introduction

The homogeneous area-preserving Hénon route naturally produces a cubic phase. A direct three-channel Kummer representation is gauge trivial; a nonfunctorial permutation survives locally but its raw prime divisor accumulates; and scalar damping repairs convergence only by inserting a new clock. These failures suggest a change in mathematical object rather than another weight adjustment.

The curve

$$E : y^2 = x^3 + 1 \tag{1}$$

is the smallest smooth projective arithmetic object carrying the same cubic symmetry. Over a field containing ζ_3 , the map $(x, y) \mapsto (\zeta_3 x, y)$ is an order-three endomorphism. The curve has $j = 0$ and complex multiplication by $\mathbb{Z}[\zeta_3]$; see [1, 2]. This paper asks what exact arithmetic data the cubic channel obtains from this completion and where the Hénon claim must stop.

We obtain a positive local theorem, a two-implementation numerical certificate, and a negative provenance conclusion. The positive theorem is intrinsic arithmetic. The negative conclusion prevents an elliptic L -function from being promoted coordinatewise to a Riemann or Hénon determinant.

2 Local Frobenius factors

For a prime $p > 3$, let

$$N_p = \#E(\mathbb{F}_p), \quad a_p = p + 1 - N_p.$$

The local zeta numerator is

$$L_p(T) = 1 - a_p T + pT^2. \tag{2}$$

Its roots have absolute value $\sqrt{p}$ by the elliptic-curve Riemann hypothesis, equivalently $|a_p| \leq 2\sqrt{p}$. This is intrinsic square-root normalization, not a fitted factor.

Theorem 2.1 (Inert-prime trace vanishing). *If $p > 3$ and $p \equiv 2 \pmod{3}$, then $a_p = 0$ and $L_p(T) = 1 + pT^2$.*

Proof. Let χ be the quadratic character of $\mathbb{F}_p$, extended by $\chi(0) = 0$. Point counting gives

$$N_p = p + 1 + \sum_{x \in \mathbb{F}_p} \chi(x^3 + 1), \quad a_p = - \sum_x \chi(x^3 + 1).$$

Because $\gcd(3, p-1) = 1$, the map $x \mapsto x^3$ is a bijection of $\mathbb{F}_p$. Therefore

$$\sum_x \chi(x^3 + 1) = \sum_u \chi(u + 1) = 0,$$

and the claim follows. □

For split primes $p \equiv 1 \pmod{3}$, the trace need not vanish. Direct counts give

$$a_7 = -4, \quad a_{13} = 2, \quad a_{19} = 8.$$

Thus the cubic channel distinguishes the inert and split prime classes without a fitted classifier.

3 Quadratic-extension independent check

Let α_p, β_p be the roots of (2). Then $\alpha_p + \beta_p = a_p$, $\alpha_p \beta_p = p$, and

$$\alpha_p^2 + \beta_p^2 = a_p^2 - 2p.$$

Hence

$$\#E(\mathbb{F}_{p^2}) = p^2 + 1 - (a_p^2 - 2p). \tag{3}$$

The release code verifies (3) independently. For each of $p = 5, 7, 11, 13$, it chooses a quadratic nonsquare d , represents $\mathbb{F}_{p^2}$ as pairs in $\mathbb{F}_p[u]/(u^2 - d)$, and tests square status for every x in the extension. This computation does not reuse the prime-field point list.

Table 1: Frozen local sentinels.				
p	$p \bmod 3$	a_p	$L_p(T)$	
5	2	0	$1 + 5T^2$	
7	1	-4	$1 + 4T + 7T^2$	
11	2	0	$1 + 11T^2$	
13	1	2	$1 - 2T + 13T^2$	
19	1	8	$1 - 8T + 19T^2$	

4 What the bridge proves

The construction solves the provenance defect of C40 at one level. The factor p in (2) and the size $\sqrt{p}$ of Frobenius are forced by geometry. The local traces are reproducible, and repetitions are controlled by the recurrence $a_{p,r} = a_p a_{p,r-1} - p a_{p,r-2}$.

It does not solve the Hénon identification. The curve (1) is selected because it carries cubic symmetry, but no theorem maps primitive Hénon unstable orbits to its Frobenius classes. Its Euler product is the elliptic L -function of E , not the Riemann zeta function. Its degree, bad factors, gamma factor, and nontrivial zeros are correspondingly different. This is a new arithmetic connection, not a Hilbert–Pólya realization.

5 Evaluator audit

Route A credits reproducible Frobenius data at A1, but only weakly because Hénon chronology is missing. The classical elliptic Euler product earns an analytic determinant label at A2. CM and square-root normalization give partial A3. Cohomology is only an A4 formal hint relative to the Hénon map. The strict tuple is

$$(A1_{\text{WEAK}}, A2_{\text{ANALYTIC DET}}, A3_{\text{PARTIAL ANALYTIC}}, A4_{\text{FORMAL HINT}}).$$

A limited Route-B audit stops at B1. No Hénon-derived Hilbert space, densely defined operator, domain, or spectral parameter map has been given. An arithmetic Euler product cannot substitute for those data.

6 Conclusion

The cubic Hénon channel has a natural arithmetic completion, and that completion is nontrivial: the $j = 0$ CM curve supplies exact inert/split Frobenius behavior and intrinsic square-root normalization. It also changes the target. The next decisive question is finite and local: can any virtual combination of this H^1 factor and Tate factors equal the Riemann local factor without inserting an exact negative copy? A three-prime comparison will answer that question before any global operator is attempted.

References

- [1] James S. Milne. *Elliptic Curves*. 2026. Author's online notes, <https://www.jmilne.org/math/Books/EC2.pdf>.
- [2] Joseph H. Silverman. *The Arithmetic of Elliptic Curves*. Springer, second edition, 2009.